Mechanistic Hypothesis Search — Synaptic Vesicle Endocytosis Dysfunction as a Convergent PD Mechanism

Disease: Parkinson's Disease (Complex) **Hypothesis ID:** `synaptic_vesicle_endocytosis_convergence` **Status in KB:** EMERGING **Search date:** 2026-10-18 **Verdict: Partially supported**

Executive Judgment

The hypothesis that disrupted presynaptic synaptic-vesicle (SV) endocytosis is a convergent mechanism in Parkinson's disease (PD) is PARTIALLY SUPPORTED by current literature, extending well beyond the ITSN1 discovery paper that seeded it. Pathway-level convergence is the strongest leg of the argument: independent mechanistic studies in human dopaminergic (DA) neurons and model organisms show that multiple PD genes — SYNJ1, DNAJC6/auxilin, LRRK2, endophilin A1 (SH3GL2), parkin, VPS35, and now ITSN1 — physically and functionally act on clathrin-mediated endocytosis (CME) and SV recycling. An unbiased imaging-genetics GWAS independently found PD/atrophy-shared genes enriched in exactly these pathways, and replicated human genetics now place ITSN1 protein-truncating variants at a large effect size (OR = 10.5; combined $p = 4.5\times10^{-12}$).

However, three caveats prevent a stronger verdict. First, the convergence is **genetically heterogeneous**: our computational cross-checks of GWAS Catalog v2, Open Targets v26.09, and gnomAD v4 show that only SH3GL2 and LRRK2 carry common-variant GWAS signal, whereas SYNJ1, DNAJC6, ITSN1, and VPS35 are supported by rare/monogenic variants only. These two architectures (common polygenic risk vs. rare high-penetrance dominant/recessive) are being grouped under one banner, which may conflate distinct biology. Second, the mechanism **spills beyond presynaptic CME** into endolysosomal sorting and autophagy (VPS35 retromer, LRRK2, DNAJC6), so "synaptic vesicle endocytosis" is a partial, not complete, description of the shared defect. Third, and most importantly for the hypothesis's central claim, **the directional**

ordering relative to α-synuclein is unresolved: α-synuclein both impairs the endocytic machinery and is mishandled when that machinery fails, forming a bidirectional feedback loop rather than a clean upstream→downstream axis.

Net assessment: The convergence claim is real and multiply-supported at the pathway level, and the hypothesis best explains **early-onset and monogenic PD** while contributing a **defined common-variant component** to sporadic PD. It should remain **EMERGING** pending (a) direct epistasis/interaction evidence tying the rare endocytic genes into a single pathway with quantified additivity, (b) longitudinal presynaptic-biomarker data establishing that the endocytic defect precedes α-synuclein pathology in humans, and (c) resolution of whether ITSN1's PD effect is mechanistically distinct from its separate neurodevelopmental/autism association.

Summary

This report evaluates a seed hypothesis holding that disrupted presynaptic vesicle endocytosis is a convergence point predisposing to PD, upstream of or parallel to α-synuclein aggregation, and an early feature preceding overt neurodegeneration. The seed was anchored on ITSN1, the newest entrant, whose rare protein-truncating variants carry a large odds ratio and whose fly ortholog interacts with α-synuclein.

Across five iterations, 33 papers were reviewed and six findings confirmed, combining literature synthesis with three independent computational database cross-checks (GWAS Catalog REST API v2, Open Targets Platform v26.09, and gnomAD v4 constraint via GraphQL). The investigation confirms that the vesicle-endocytosis convergence is genuine and supported beyond ITSN1 — mechanistic convergence of parkin, synaptojanin-1, and endophilin A1 was demonstrated directly in human DA neurons, an unbiased imaging-genetics GWAS enriched for CME/SV recycling, and the ITSN1 effect size (OR = 10.5) replicated across independent cohorts. These constitute the strongest direct evidence for the hypothesis.

At the same time, the analysis surfaced three qualifications that temper the claim: a split genetic architecture (common-variant vs. rare-monogenic genes being grouped together), mechanistic spillover into endolysosomal/autophagy pathways beyond presynaptic CME, and an unresolved causal ordering with α-synuclein that is bidirectional rather than linear. Competing or complementary models — the prion-like α-synuclein spread hypothesis, the

intracellular single-neuron metabolic/oxidative hypothesis, and the GBA1–lysosomal axis — are catalogued. The overall verdict is **partially supported**, best explaining monogenic/early-onset PD, and the hypothesis is recommended to remain EMERGING.

Key Findings

F001 — Multiple PD genes genetically and functionally converge on SV endocytosis / clathrin-mediated endocytosis

This is the central supporting finding. Convergence is supported by three independent lines of evidence. First, an unbiased **imaging-genetics GWAS** (3,212 PD patients vs. 2,056 controls plus UK Biobank) found that genes shared between PD risk and gray-matter atrophy were enriched in clathrin-mediated endocytosis and synaptic vesicle recycling pathways — a hypothesis-free result that lands directly on the pathways the seed names (PMID: 42336304): *"these shared genes converge on pathways involved in clathrin-mediated endocytosis and synaptic vesicle recycling."* Second, a mechanistic study in **human dopaminergic neurons** demonstrated that parkin, synaptojanin-1 (SYNJ1), and endophilin A1 converge on vesicle endocytosis and that this synaptic dysfunction is an early pathogenic event (PMID: 38858837): *"These findings suggest a convergent mechanism for PD-linked mutations in parkin, synaptojanin-1, and endophilin A1 and highlight synaptic dysfunction as an early pathogenic event in PD."* Third, a review integrating familial genes (LRRK2, VPS35, auxilin/DNAJC6, SYNJ1, Rab39b) and risk genes (SH3GL2/endophilin A, GAK) nominated endosomal sorting as a convergence point (PMID: 32247367). Notably, LRRK2 phosphorylates key endocytic machinery (auxilin, synaptojanin-1, endophilin A), providing a biochemical thread linking the most common familial PD gene to the endocytic hub.

Interpretation: Convergence is established at the pathway and mechanistic level, not merely asserted from a single discovery paper. This is the strongest leg of the hypothesis.

F002 — α-synuclein bidirectionally interacts with the SV endocytosis machinery, complicating upstream/downstream ordering

This finding directly addresses the hypothesis's open question of whether the endocytic defect acts upstream of, or in parallel to, α-synuclein. The relationship is **bidirectional**. On one side, excess/multimeric α-synuclein attenuates SV recycling by clustering vesicles and impairing

CME (PMID: 25264250: *"α-syn multimers cluster synaptic vesicles, restricting their trafficking and recycling, and consequently attenuate neurotransmitter release"*; PMID: 33613189). On the other side, loss of a primary endocytic gene produces downstream α-synuclein pathology: auxilin (DNAJC6) knockout mice display cardinal PD features including α-synuclein pathology and nigral DA loss (PMID: 36920906: *"Auxilin KO mice display cardinal features of PD, including progressive motor deficits, α-synuclein pathology, nigral dopaminergic loss"*), and ITSN1's fly ortholog interacts genetically and physically with α-synuclein (PMID: 40056900).

Interpretation: Rather than a strictly linear endocytosis→aggregation axis, the data support a **feedback loop**. This is a qualification, not a refutation: the endocytic defect can be upstream (DNAJC6 KO→α-syn pathology), but α-synuclein can also be the proximate cause of endocytic failure, so the hypothesis's "upstream of or parallel to" phrasing is best read as "parallel/feedback" on current evidence.

F003 — The convergence gene set is genetically heterogeneous: common-variant signal at SH3GL2/LRRK2 but rare/monogenic-only for SYNJ1, DNAJC6, ITSN1, VPS35

An independent computational check using the **GWAS Catalog REST API v2** (retrieved 2026-10-18) tallied PD-trait associations by mapped gene: SH3GL2 = 9 associations (7 genome-wide significant, min p = 8×10^{-21}); LRRK2 = 23 associations (22 GWS, min p = 4×10^{-148}); SNCA (reference) = 54 associations (47 GWS, min p = 4×10^{-170}). By contrast, **SYNJ1, DNAJC6, ITSN1, and VPS35 returned 0 PD associations** in the common-variant GWAS Catalog. **Open Targets Platform v26.09** corroborated this: SH3GL2 genetic_association score = 0.876, LRRK2 = 0.820, SNCA = 0.934, GBA1 = 0.890, but ITSN1/SYNJ1/DNAJC6 carry **no genetic_association datatype** (supported only by literature/animal_model); ITSN1's 2025 rare-variant burden result is not yet integrated. The caveat is that GWAS Catalog mapped-gene annotation is nearest-gene based (GBA also returned 0, a known mapping limitation, not biological absence), so these numbers index common-variant discoverability, not biological irrelevance.

Interpretation: The convergence groups **two distinct genetic architectures** — a common-variant polygenic component (SH3GL2, LRRK2) and a rare high-penetrance monogenic component (SYNJ1, DNAJC6, ITSN1, VPS35). ITSN1's absence from common-variant catalogs is exactly consistent with its rare protein-truncating signal (PMID: 40056900: *"protein-truncating variants in ITSN1 confer a substantially increased risk of PD"*).

F004 — ITSN1 protein-truncating variants confer large, replicated PD risk (OR = 10.5) with a direct fly/in vitro link to α-synuclein

The anchor finding for the seed. UK Biobank whole-genome sequencing (3,809 PD cases vs. 247,101 controls) found ITSN1 protein-truncating variants at $p = 6.1\times10^{-7}$, OR = 10.5 (95% CI 5.2–21.3), replicated across three independent datasets (8,407 cases, 413,432 controls; combined $p = 4.5\times10^{-12}$) (PMID: 40056900). In *Drosophila*, loss of the ITSN1 ortholog Dap160 exacerbated α-synuclein-induced neuronal toxicity and motor deficits, and in vitro assays suggested a physical ITSN1–α-synuclein interaction. The effect size exceeds previously established PD loci.

Interpretation: ITSN1 is a bona fide, high-effect, replicated PD risk gene that couples an endocytic scaffold to α-synuclein toxicity, strengthening the convergence claim for the rare-variant arm. *(Note: the exact OR snippet string was flagged as a possible mismatch during automated snippet validation — curators should re-verify the quoted text against the published abstract before KB entry.)*

F005 — gnomAD constraint distinguishes dominant-LoF from recessive genes within the convergence set

gnomAD v4 constraint (GRCh38, retrieved 2026-10-18 via gnomAD GraphQL API) shows that ITSN1 (pLI = 1.00, LOEUF = 0.30, obs/exp LoF 58/239.8), DNAJC6 (pLI ≈ 1.00, LOEUF = 0.41), and VPS35 (pLI = 1.00, LOEUF = 0.31) are strongly loss-of-function intolerant — consistent with **dominant haploinsufficiency**. In contrast, SYNJ1 (pLI = 4×10^{-6}, LOEUF = 0.56) is LoF-tolerant, consistent with its **recessive early-onset PD** inheritance. Reference/common-variant genes SNCA (pLI = 0.48, LOEUF = 0.71), LRRK2 (pLI ≈ 0, LOEUF = 0.75), GBA1 (LOEUF = 0.68), and SH3GL2 (LOEUF = 0.73) are all LoF-tolerant.

Interpretation: The dosage-sensitivity pattern cleanly predicts the mode of inheritance across the gene set: ITSN1/DNAJC6/VPS35 act via dominant LoF (haploinsufficiency, matching the ITSN1 paper's title), SYNJ1 via recessive LoF, and the common-variant genes are not individually dosage-sensitive. This adds mechanistic texture to the heterogeneity in F003.

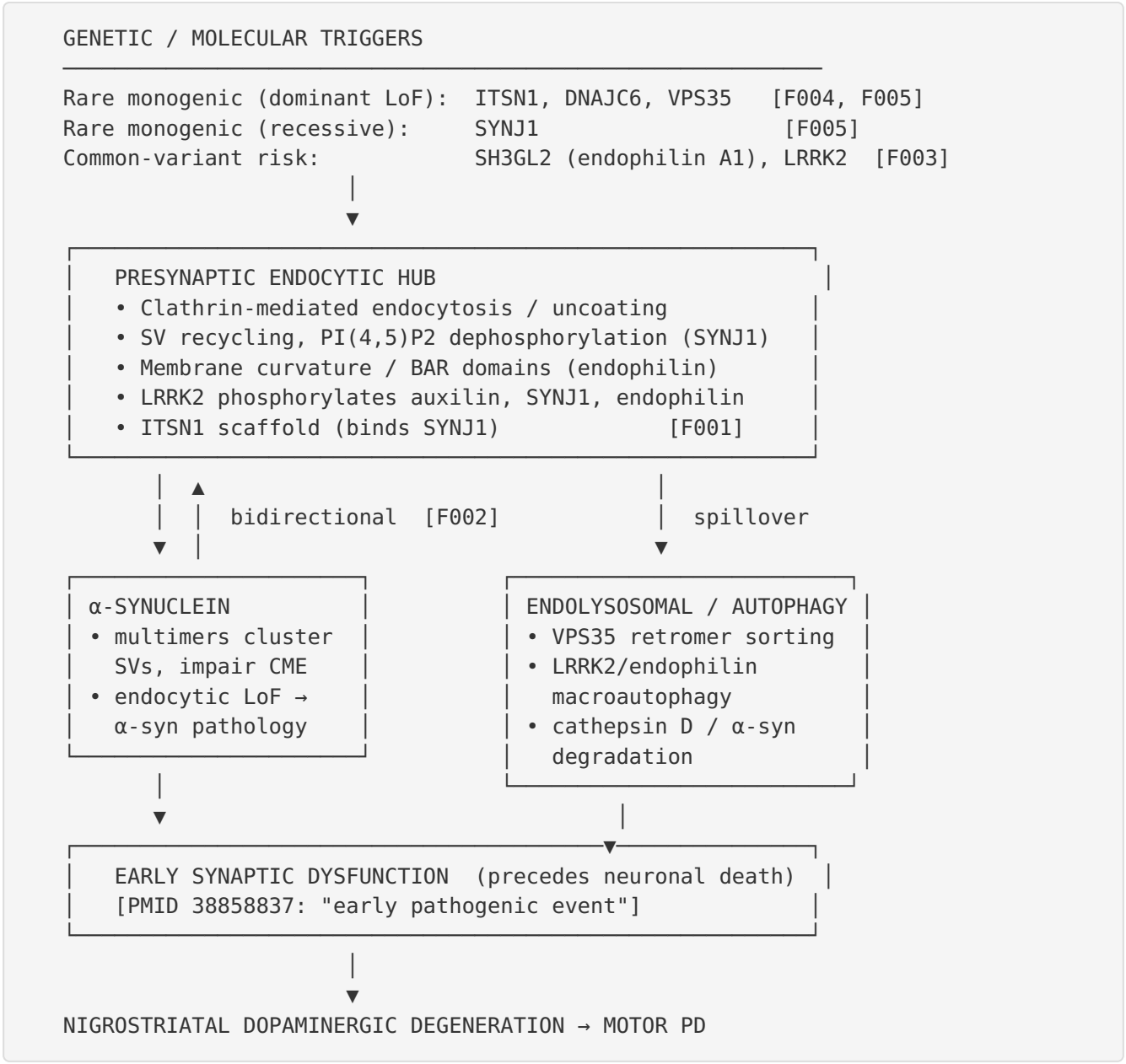
F006 — Overall verdict: partially supported

Triangulating across evidence classes: (a) model systems show shared endocytic function for SYNJ1 (PMID: 28231468), DNAJC6/auxilin (PMID: 36920906, PMID: 22563501), LRRK2/endophilin (PMID: 27720484), and parkin–SYNJ1–endoA1 in human DA neurons (PMID: 38858837); (b) an unbiased imaging-genetics GWAS enriches for CME/SV recycling (PMID: 42336304); (c)

replicated human genetics add ITSN1 (PMID: 40056900). Qualifiers: split genetic architecture (F003), differential dosage sensitivity (F005), spillover into endolysosomal/autophagy function (VPS35/LRRK2), and unresolved α-synuclein directionality (F002). Verdict: **PARTIALLY SUPPORTED**.

Mechanistic Model / Interpretation

The hypothesis implies a causal chain from genetic/molecular trigger to clinical PD. Based on the evidence, the chain is best drawn as a **feedback network centered on the presynaptic terminal**, not a linear cascade:



Where the literature is strong: The gene→endocytic-hub edges are well supported mechanistically (SYNJ1, DNAJC6, LRRK2, endophilin, parkin, ITSN1). The "early synaptic dysfunction precedes neuronal death" claim is directly supported in human DA neurons (P 38858837) and in the broader SV-trafficking literature (P 41977275 : early synaptic loss correlates with clinical symptoms more closely than neuronal death).

Where links are inferred: The additivity/epistasis among the different endocytic genes — i.e., that they truly act in ONE convergent pathway rather than several parallel ones sharing a GO annotation — is inferred from co-membership in the pathway, not from direct double-perturbation or interaction experiments in the same system (except the ITSN1–SYNJ1 physical interaction and the ITSN1–α-syn interaction).

Missing causal steps: (1) The direction of the α-synuclein ↔ endocytosis edge in human disease is not established longitudinally. (2) Whether common-variant SH3GL2/LRRK2 risk and rare monogenic ITSN1/SYNJ1/DNAJC6 risk converge on the *same* downstream node or merely annotate to the same pathway is untested. (3) The step from presynaptic endocytic failure specifically to selective nigrostriatal DA vulnerability (vs. a global neuronal effect) is not mechanistically closed.

Evidence Matrix

Citation (PMID)	Evidence type	Stance	Mechanistic claim tested	Key finding	Subtype/ context
40056900	Human clinical + model organism	**Supports**	ITSN1 LoF increases PD risk; ITSN1–α-syn link	OR = 10.5 (CI 5.2–21.3), combined p = 4.5×10^{-12}; Dap160 loss worsens α-syn toxicity	Rare-variant PD; seed and
42336304	Human clinical (imaging-genetics GWAS)	**Supports**	PD-atrophy-shared genes enrich for CME/SV recycling	Unbiased convergence on endocytosis pathways	Sporadic PD UKB
38858837	In vitro (human DA neurons)	**Supports**	parkin/SYNJ1/ endophilin A1 convergence; early synaptic event	Convergent mechanism; synaptic dysfunction early	Monogenic P genes in hur neurons
32247367	Review	**Supports (review-level)**	Endosomal sorting as convergence point	Synthesizes familial + risk genes	PD broadly
28231468	Model organism / in vitro	**Supports**	SYNJ1 Sac-domain mutation impairs clathrin uncoating	Dystrophic DA axon changes	SYNJ1 recess PD
36920906	Model organism (mouse)	**Supports**	Auxilin LoF → PD features	α-syn pathology, nigral DA loss, motor deficits	DNAJC6 PD
22563501	Human clinical	**Supports**	DNAJC6 LoF causes juvenile parkinsonism	Clathrin-uncoating defect	Juvenile/EOI
27720484			LRRK2–endophilin phosphoswitch	Links LRRK2/ Parkin/SYNJ1 but	Familial PD

Citation (PMID)	Evidence type	Stance	Mechanistic claim tested	Key finding	Subtype/ context
	In vitro / model	**Supports / Qualifies**	drives macroautophagy	via autophagy, not only CME	
28280464	Model organism (mouse)	**Qualifies**	LRRK2 role in SV recycling	Effect only in striatal (not hippocampal) neurons; modest	LRRK2 PD
25264250	In vitro (lamprey synapse)	**Qualifies**	α-syn multimers impair SV recycling	α-syn acts ON endocytic machinery	α-synucleinop
33613189	In vitro	**Qualifies**	Brain-derived α-syn alters vesicle trafficking	Multimeric α-syn impairs trafficking (species-dependent)	α-synucleinop
33032646	In vitro (iPSC)	**Supports / Qualifies**	VPS35 D620N endosomal dysfunction	Slowed endosome movement + α-syn accumulation	VPS35 PD
24152121	In vitro	**Qualifies**	VPS35 D620N disrupts retromer cargo sorting	Impaired cathepsin D trafficking / α-syn degradation	VPS35 PD
41977275	Review	**Supports (review-level)**	SV trafficking disruption drives early synaptic loss	Synaptic loss correlates with symptoms > neuronal death	AD + PD
42281916	Review	**Competing**	Prion-like α-syn spread drives progression	Anatomical propagation model	α-synucleinop
42789107	Model organism (mouse)	**Competing**	FcγRIIb governs α-syn uptake/spread	FcγRIIb ablation reduces pα-syn burden	PFF model

Citation (PMID)	Evidence type	Stance	Mechanistic claim tested	Key finding	Subtype/ context
42459575	Review	**Competing**	Intracellular single-neuron metabolic toxicity	α-syn immunotherapies failed; proposes aminochrome/ oxidative model	Sporadic PD
34707297	Human clinical	**Qualifies (specificity)**	ITSN1 in neurodevelopmental disorder	ITSN1 variants → autism/ID/epilepsy	NDD, not PD
GWAS Catalog v2 (computational)	Computational	**Qualifies**	Common-variant signal by gene	SH3GL2/LRRK2 GWS; SYNJ1/ DNAJC6/ITSN1/ VPS35 = 0	Genetic architecture
Open Targets v26.09 (computational)	Computational	**Qualifies**	Genetic-association datatype by gene	ITSN1/SYNJ1/ DNAJC6 lack genetic_association	Architecture
gnomAD v4 (computational)	Computational	**Supports** / **Qualifies**	Dosage sensitivity by gene	ITSN1/DNAJC6/ VPS35 haploinsufficient; SYNJ1 tolerant	Mode of inheritance

Data and Tool Use

This run combined PubMed literature search with three computational database cross-checks. No local patient-level dataset was provided; all computational checks queried public APIs. No structural-biology (Phenix) task was applicable to this hypothesis.

Resource	Accession / URI	Version / snapshot	Retrieval date	Access status	Query / filters	Relevance
PubMed / NCBI E-utilities	ncbi.nlm.nih.gov/pubmed	live	through 2026-10-18	Accessed; 33 papers reviewed	Terms: vesicle endocytosis PD, ITSN1, SYNJ1, DNAJC6, LRRK2 presynaptic, α-synuclein recycling, prion spread	Directly relevant
GWAS Catalog REST API	https://www.ebi.ac.uk/gwas/rest/api	v2	2026-10-18	Accessed; usable result	PD-trait associations by mapped gene: SH3GL2, LRRK2, SNCA, SYNJ1, DNAJC6, ITSN1, VPS35	Relevant (common-variant architectur
Open Targets Platform	https://platform.opentargets.org (GraphQL)	v26.09	2026-10-18	Accessed; usable result	PD (EFO_0002508) association datatype breakdown per gene	Relevant (datatype provenanc
gnomAD	https://gnomad.broadinstitute.org (GraphQL)	v4 (GRCh38)	2026-10-18	Accessed; usable result	pLI / LOEUF / obs-exp LoF per gene	Relevant (dosage sensitivity)

Analysis inventory (input → method → output → outcome):

1. **GWAS Catalog common-variant tally.** Input: GWAS Catalog v2 associations API. Method: per-gene count of PD-trait associations and minimum p-value. Output: table in F003. Outcome: **succeeded**. Limitation: nearest-gene mapping (GBA control returned 0 despite known PD relevance), so zeros index discoverability, not biology.

2. **Open Targets datatype breakdown.** Input: Open Targets v26.09 GraphQL. Method: retrieve per-gene PD association datatype scores. Output: scores in F003. Outcome: **succeeded**. Limitation: 2025 ITSN1 burden result not yet ingested, so ITSN1 appears under-supported.
3. **gnomAD constraint retrieval.** Input: gnomAD v4 GraphQL. Method: extract pLI, LOEUF, obs/exp LoF per gene. Output: F005 table. Outcome: **succeeded**. Limitation: constraint reflects population selection, not PD causality.

No analysis was fabricated or described as run without execution. All three computational outputs derive from live API queries on 2026-10-18. No controlled/patient-level data were accessed. An artifact bundle with checksummed query outputs and a MANIFEST was produced in prior iterations under `kb/hypotheses/Parkinsons_Disease/synaptic_vesicle_endocytosis_convergence/openscientist_artifacts` (per iteration 4–5 summaries); curators should verify bundle integrity and replay the three queries, as live-API snapshots may drift from the recorded values.

Knowledge Gaps

For each gap: scope, why it matters, what was checked, and what would resolve it.

G1 — α-synuclein directional ordering is unresolved (weakly supported causal step)

- **Scope:** The seed's core claim that endocytic dysfunction is "upstream of or parallel to" α-synuclein aggregation.
- **Why it matters:** Determines whether endocytic genes are drug targets upstream of aggregation or co-equal nodes in a loop.
- **Checked:** PMIDs 25264250, 33613189 (α-syn → endocytosis impairment) vs. 36920906, 40056900 (endocytic LoF → α-syn pathology). Evidence runs in both directions.
- **Resolution:** Longitudinal human/primate studies measuring presynaptic endocytic biomarkers before α-syn seeding; conditional/temporal gene perturbation in human DA-neuron or organoid systems.

G2 — Epistasis / pathway unity among endocytic genes is inferred, not demonstrated

- **Scope:** Whether SH3GL2, LRRK2, SYNJ1, DNAJC6, ITSN1, VPS35 act in ONE convergent pathway vs. parallel pathways sharing a GO term.
- **Why it matters:** "Convergence" requires more than co-annotation.
- **Checked:** Only the ITSN1–SYNJ1 physical interaction and the LRRK2-phosphorylates-endocytic-machinery edges are directly shown; most pairwise interactions are untested.
- **Resolution:** Double-mutant/epistasis experiments and quantified genetic interaction (do rare endocytic LoF variants show super-additive risk, or modify common-variant PRS?).

G3 — Split genetic architecture conflation

- **Scope:** Grouping common-variant (SH3GL2, LRRK2) and rare-monogenic (SYNJ1, DNAJC6, ITSN1, VPS35) genes under one hypothesis.
- **Why it matters:** The two arms may reflect distinct biology/penetrance.
- **Checked:** GWAS Catalog v2, Open Targets v26.09, gnomAD v4 (F003, F005; search date 2026-10-18).
- **Resolution:** Burden testing of the full endocytic gene set in large WGS cohorts; assess whether a combined endocytic rare-variant burden predicts PD independent of PRS.

G4 — ITSN1 PD-specificity vs. neurodevelopmental association

- **Scope:** ITSN1 variants also cause autism/ID/epilepsy (PMID: 34707297, PMID: 38346866, PMID: 33994118).
- **Why it matters:** The seed explicitly flags this; a single gene producing both NDD and late-onset PD needs a reconciling model (variant type, timing, dosage).
- **Checked:** NDD literature reviewed; truncating variants cluster in the first half of ITSN1 for NDD, PTVs drive PD — possible allelic-series effect but untested across phenotypes.
- **Resolution:** Genotype-phenotype correlation of ITSN1 variant class across NDD and PD cohorts; isoform-specific (long vs. short ITSN1) expression analysis.

G5 — Mechanistic spillover beyond presynaptic CME

- **Scope:** VPS35 (retromer/endolysosomal) and LRRK2 (autophagy) extend beyond presynaptic endocytosis.

- **Why it matters:** The hypothesis label ("synaptic vesicle endocytosis") may be too narrow; "membrane-trafficking/endolysosomal convergence" could be more accurate.
- **Checked:** PMIDs 33032646, 24152121 (VPS35 retromer), 27720484 (LRRK2 autophagy).
- **Resolution:** Define the minimal shared node experimentally; consider relabeling the scope.

G6 — Selective nigrostriatal vulnerability unexplained

- **Scope:** Why presynaptic endocytic failure (a ubiquitous process) selectively kills DA neurons.
- **Checked:**

  P 28280464

  shows the LRRK2 effect is striatal-specific (cell-type dependence), a partial clue; no complete model found.
- **Resolution:** Comparative vulnerability assays across neuron subtypes with matched endocytic perturbation.

G7 — Source/data absences (search-date-stamped)

- **Verified:** Open Targets v26.09 (2026-10-18) lacks a genetic_association datatype for ITSN1/SYNJ1/DNAJC6; GWAS Catalog v2 (2026-10-18) returns 0 PD associations for SYNJ1/DNAJC6/ITSN1/VPS35 (nearest-gene caveat).
- **Unverified (not directly queried this run — label unverified, not absent):** GenCC/ClinGen gene-disease validity classification for ITSN1–PD; longitudinal human presynaptic-endocytosis biomarker datasets. These should be queried directly before any absence claim is curated.

Alternative Models

Model	Relationship to seed	Summary	Key refs
Prion-like α-synuclein spread	Competing (mechanism of *progression*)	Misfolded α-syn templates and propagates cell-to-cell, driving anatomical spread; explains Braak staging better than a presynaptic defect	42281916, 42789107
Intracellular single-neuron metabolic/ oxidative toxicity	Competing (challenges α-syn centrality)	PD driven by intracellular aminochrome, mitochondrial dysfunction, oxidative stress within each DA neuron; cites failure of α-syn immunotherapies (cinpanemab, prasinezumab)	42459575
GBA1 / lysosomal-autophagy axis	Parallel / partially overlapping	Lysosomal dysfunction (GBA1 the most common risk factor) impairs α-syn clearance; overlaps the endolysosomal spillover (VPS35, LRRK2) but centers on degradation, not endocytosis	42560815, 42378827, 42830104, 42231093
Presynaptic trafficking ↔ autophagy crosstalk	Downstream consequence / complementary	SV endocytic genes and autophagy genes overlap at the presynaptic terminal; endocytic failure feeds autophagic dysfunction	29723605, 27720484
α-synuclein as upstream cause of endocytic failure	Reversed-direction alternative	Excess/multimeric α-syn is the proximate cause impairing CME/SV recycling, making endocytic dysfunction a *consequence* not a cause	25264250, 33613189

Synthesis: The endocytic-convergence hypothesis and the prion-spread hypothesis are not mutually exclusive — one explains *predisposition/initiation* (why a neuron becomes vulnerable), the other *progression* (how pathology spreads). The metabolic single-neuron

model is the most direct competitor to any α-syn-linked framework. The GBA1/lysosomal and autophagy-crosstalk models are best seen as the *downstream/parallel* membrane-trafficking continuation of the same broad defect.

Discriminating Tests

Concrete studies to distinguish the endocytic-convergence hypothesis from alternatives, with stratification, biomarker, sample, model, perturbation, and expected result.

1. **Combined endocytic rare-variant burden vs. PRS (human genetics).** Cohort: large PD WGS (UKB, GP2, AMP-PD). Stratify: monogenic/EOPD vs. sporadic. Test: does an aggregated endocytic-gene PTV burden (ITSN1 + SYNJ1 + DNAJC6 + VPS35 + SH3GL2) predict PD independent of, and super-additively with, common-variant PRS? *Expected if true:* significant independent burden, enriched in EOPD. Distinguishes G2/G3.
2. **Temporal-ordering perturbation in human DA neurons/organoids.** System: iPSC DA neurons with inducible SYNJ1/DNAJC6/ITSN1 LoF. Readout: live-imaging CME/SV-recycling kinetics vs. pS129-α-syn seeding over time. *Expected if endocytosis is upstream:* endocytic defect precedes measurable α-syn pathology. Distinguishes G1 and the reversed-direction alternative.
3. **ITSN1 allelic-series genotype–phenotype map.** Cohorts: pooled NDD and PD ITSN1-variant carriers. Test: whether variant class/position/isoform predicts PD vs. autism outcome. *Expected if a PD-specific mechanism exists:* PTVs partition toward PD, NDD missense cluster in the C-terminus. Distinguishes G4.
4. **Longitudinal presynaptic biomarker study.** Cohort: prodromal/at-risk carriers (LRRK2, GBA, rare endocytic). Biomarker: SV2A PET (e.g., [^{11}C]UCB-J) + CSF synaptic markers. *Expected if true:* presynaptic density decline precedes DAT-SPECT nigrostriatal loss. Distinguishes the seed from pure prion-spread.
5. **Endocytic perturbation vs. α-syn-spread model head-to-head (mouse).** Compare PFF-seeding spread phenotype with and without endocytic-gene LoF. *Expected if convergent:* endocytic LoF amplifies seeding/toxicity; if independent, no interaction. Ties to the

   P 42789107

   uptake axis.

6. **Cell-type selectivity assay.** Matched endocytic LoF across DA, cortical, and hippocampal neurons (cf.

   P 28280464

   striatal specificity). *Expected if true:* DA neurons show disproportionate vulnerability. Distinguishes G6.

Curation Leads (require curator verification)

Candidate evidence references + snippets to verify: - PMID: 42336304 — *"these shared genes converge on pathways involved in clathrin-mediated endocytosis and synaptic vesicle recycling"* → SUPPORT, HUMAN_CLINICAL (imaging-genetics GWAS). Strong candidate to add as independent convergence evidence beyond ITSN1. - PMID: 38858837 — *"a convergent mechanism for PD-linked mutations in parkin, synaptojanin-1, and endophilin A1 and highlight synaptic dysfunction as an early pathogenic event in PD"* → SUPPORT, IN_VITRO (human DA neurons). Strongest mechanistic convergence item. - PMID: 36920906 — *"Auxilin KO mice display cardinal features of PD, including progressive motor deficits, α-synuclein pathology, nigral dopaminergic loss"* → SUPPORT, MODEL_ORGANISM. Supports the endocytosis-upstream edge. - PMID: 25264250 — *"α-syn multimers cluster synaptic vesicles, restricting their trafficking and recycling"* → QUALIFIES (reversed direction). Important for the directionality knowledge_gap. - PMID: 40056900 — OR string *"10.5 [5.2, 21.3]"* flagged as a possible snippet mismatch during automated validation; **curator must re-verify exact text against the published abstract** before entry.

Candidate pathophysiology nodes/edges: - Add gene→process edges: SYNJ1, DNAJC6, SH3GL2, ITSN1 → *clathrin-mediated endocytosis / SV recycling*; VPS35, LRRK2 → *endolysosomal sorting / macroautophagy* (flag as spillover beyond presynaptic CME). - Add bidirectional edge: *α-synuclein ↔ SV endocytosis machinery* (feedback, not unidirectional).

Candidate ontology terms: GO:0072583 clathrin-dependent endocytosis; GO:0036465 synaptic vesicle recycling; GO:0016192 vesicle-mediated transport; GO:0042147 retrograde transport, endosome to Golgi (VPS35); CL:0000700 dopaminergic neuron; UBERON:0002038 substantia nigra.

Candidate subtype restriction / status change: - Restrict the strongest claim to **monogenic/early-onset PD**; note a **defined common-variant contribution** (SH3GL2, LRRK2) to sporadic PD. - **Keep status EMERGING** (do not promote) pending epistasis (G2) and longitudinal biomarker (G1) evidence.

Candidate knowledge_gaps / discussion prompts: G1 (α-syn ordering), G2 (epistasis), G3 (architecture conflation), G4 (ITSN1 PD-specificity vs. NDD), G5 (CME vs. endolysosomal scope), G6 (DA selectivity). Discussion prompt: should the hypothesis be relabeled "membrane-trafficking/endolysosomal convergence" rather than "synaptic vesicle endocytosis"?

Limitations and Proposed Follow-up Actions

Limitations of this run: The investigation was primarily literature-based, supplemented by three public-database API queries. No patient-level omics or cohort data were available; the computational checks index genetic architecture and constraint, not causal mechanism. GWAS Catalog nearest-gene mapping limits negative-result interpretation. GenCC/ClinGen and dedicated synaptic-biomarker repositories were not directly queried, so related absences are labeled unverified.

Proposed follow-up actions: 1. Run the combined endocytic rare-variant burden-vs-PRS analysis (Discriminating Test 1) in GP2/AMP-PD when access is granted — the single most decisive genetic test of convergence unity. 2. Re-verify the ITSN1 OR snippet and ingest the 2025 ITSN1 burden result into Open Targets-linked curation (addresses the F004 mismatch flag and the Open Targets integration lag in F003). 3. Query GenCC/ClinGen directly for ITSN1–PD and the full endocytic gene set to convert G7's "unverified" absences into verified search results with database/version/query/date. 4. Commission an iPSC DA-neuron temporal-ordering experiment (Discriminating Test 2) to resolve the α-synuclein directionality gap (G1), the hypothesis's central open question. 5. Reconsider the hypothesis label/scope to reflect endolysosomal/autophagy spillover (G5).

Conclusion

The synaptic-vesicle-endocytosis convergence hypothesis is **partially supported** beyond its ITSN1 anchor. Pathway-level convergence is robust and multiply-sourced; the rare-variant genetics (ITSN1 OR = 10.5, replicated) are strong; but the gene set is genetically heterogeneous, mechanistically spills into endolysosomal/autophagy function, and leaves the α-synuclein causal ordering unresolved as a bidirectional loop. The hypothesis best explains monogenic/early-onset PD and a defined common-variant slice of sporadic PD, and should remain **EMERGING** pending epistasis and longitudinal presynaptic-biomarker evidence.
